

Clarkson Institute Summer School: Examples Sheet 4 - Product groups

Please send all comments and corrections to jmm232@cam.ac.uk.

Direct products

1. (a) Construct the group table of $C_2 \times C_3$. Find the orders of all the elements, and hence verify the isomorphism $C_2 \times C_3 \cong C_6$.
- (b) Construct the group table of $C_3 \times C_3$. Find the orders of all the elements, and hence verify that $C_3 \times C_3 \not\cong C_9$.

Solution:

- (a) Let $C_2 = \{e, r\}$ and $C_3 = \{e, r, r^2\}$. The elements of $C_2 \times C_3$ are then:

$$C_2 \times C_3 = \{(e, e), (e, r), (e, r^2), (r, e), (r, r), (r, r^2)\}.$$

The group table is given by:

*	(e, e)	(e, r)	(e, r^2)	(r, e)	(r, r)	(r, r^2)
(e, e)	(e, e)	(e, r)	(e, r^2)	(r, e)	(r, r)	(r, r^2)
(e, r)	(e, r)	(e, r^2)	(e, e)	(r, r)	(r, r^2)	(r, e)
(e, r^2)	(e, r^2)	(e, e)	(e, r)	(r, r^2)	(r, e)	(r, r)
(r, e)	(r, e)	(r, r)	(r, r^2)	(e, e)	(e, r)	(e, r^2)
(r, r)	(r, r)	(r, r^2)	(r, e)	(e, r)	(e, r^2)	(e, e)
(r, r^2)	(r, r^2)	(r, e)	(r, r)	(e, r^2)	(e, e)	(e, r)

The order of (e, e) is one, the order of (e, r) is three, the order of (e, r^2) is three, the order of (r, e) is two, the order of (r, r) is six, and the order of (r, r^2) is six. Since the group contains an element of order six, it must be cyclic.

- (b) Now let $C_3 = \{e, r, r^2\}$. The elements of $C_3 \times C_3$ are given by:

$$C_3 \times C_3 = \{(e, e), (e, r), (e, r^2), (r, e), (r, r), (r, r^2), (r^2, e), (r^2, r), (r^2, r^2)\}.$$

The group table is given by:

*	(e, e)	(e, r)	(e, r^2)	(r, e)	(r, r)	(r, r^2)	(r^2, e)	(r^2, r)	(r^2, r^2)
(e, e)	(e, e)	(e, r)	(e, r^2)	(r, e)	(r, r)	(r, r^2)	(r^2, e)	(r^2, r)	(r^2, r^2)
(e, r)	(e, r)	(e, r^2)	(e, e)	(r, r)	(r, r^2)	(r, e)	(r^2, r)	(r^2, r^2)	(r^2, e)
(e, r^2)	(e, r^2)	(e, e)	(e, r)	(r, r^2)	(r, e)	(r, r)	(r^2, r^2)	(r^2, e)	(r^2, r)
(r, e)	(r, e)	(r, r)	(r, r^2)	(r^2, e)	(r^2, r)	(r^2, r^2)	(e, e)	(e, r)	(e, r^2)
(r, r)	(r, r)	(r, r^2)	(r, e)	(r^2, r)	(r^2, r^2)	(r^2, e)	(e, r)	(e, r^2)	(e, e)
(r, r^2)	(r, r^2)	(r, e)	(r, r)	(r^2, r^2)	(r^2, e)	(r^2, r)	(e, r^2)	(e, e)	(e, r)
(r^2, e)	(r^2, e)	(r^2, r)	(r^2, r^2)	(e, e)	(e, r)	(e, r^2)	(r, e)	(r, r)	(r, r^2)
(r^2, r)	(r^2, r)	(r^2, r^2)	(r^2, e)	(e, r)	(e, r^2)	(e, e)	(r, r)	(r, r^2)	(r, e)
(r^2, r^2)	(r^2, r^2)	(r^2, e)	(r^2, r)	(e, r^2)	(e, e)	(e, r)	(r, r^2)	(r, e)	(r, r)

The orders of the elements are: (e, e) has order one, (e, r) has order three, (e, r^2) has order three, (r, e) has order three, (r, r) has order three, (r, r^2) has order three, (r^2, e) has order three, (r^2, r) has order three, and (r^2, r^2) has order three. Since this group does not contain an element of order nine, it cannot be isomorphic to C_9 .

2. Is D_3 isomorphic to the direct product of two non-trivial groups? Exhibit two such groups, or show that none can exist.

Solution: Suppose that D_3 is isomorphic to some product $H \times K$. Then considering sizes, we would need $|H||K| = 6$. But then one of H, K would have size two, and the other would have size three, since they are non-trivial. But this cannot be the case, because then H, K would both be Abelian (they are isomorphic to C_2, C_3 respectively), and hence $H \times K$ would be Abelian, and hence could not be isomorphic to D_3 . Thus D_3 is not isomorphic to a direct product of non-trivial groups.

3. Show that for any three groups G_1, G_2, G_3 , we have the following:

- (a) $G_1 \times G_2 \cong G_2 \times G_1$ (hence, the direct product is 'commutative', up to isomorphism);
 (b) $G_1 \times (G_2 \times G_3) \cong (G_1 \times G_2) \times G_3$ (hence, the direct product is 'associative', up to isomorphism).

Thus, we may be quite relaxed when manipulating direct products - e.g. it is perfectly reasonable to write $C_2 \times C_2 \times C_3 \cong C_2 \times C_3 \times C_2$ without thinking too much.

Solution: We construct the isomorphisms in each case:

- (a) In the first case, define $f : G_1 \times G_2 \rightarrow G_2 \times G_1$ via $f(g_1, g_2) = (g_2, g_1)$. This is obviously invertible (with inverse $f^{-1}(g_2, g_1) = (g_1, g_2)$), and satisfies the requirement that we preserve the group multiplication:

$$f((g_1, g_2)(g'_1, g'_2)) = f(g_1g'_1, g_2g'_2) = (g_2g'_2, g_1g'_1) = (g_2, g_1)(g'_2, g'_1) = f(g_1, g_2)f(g'_1, g'_2).$$

Hence, this is an isomorphism, as required.

- (b) In the second case, define $f : G_1 \times (G_2 \times G_3) \rightarrow (G_1 \times G_2) \times G_3$ via:

$$f(g_1, (g_2, g_3)) = ((g_1, g_2), g_3).$$

Again, this is obviously invertible (with inverse $f^{-1}((g_1, g_2), g_3) = (g_1, (g_2, g_3))$), and satisfies the requirement that we preserve the group multiplication, since we have:

$$\begin{aligned} f([g_1, (g_2, g_3)][g'_1, (g'_2, g'_3)]) &= f(g_1g'_1, (g_2, g_3)(g'_2, g'_3)) = f(g_1g'_1, (g_2g'_2, g_3g'_3)) \\ &= ((g_1g'_1, g_2g'_2), g_3g'_3), \end{aligned}$$

and also:

$$\begin{aligned} f(g_1, (g_2, g_3))f(g'_1, (g'_2, g'_3)) &= ((g_1, g_2), g_3)((g'_1, g'_2), g'_3) = ((g_1, g_2)(g'_1, g'_2), g_3g'_3) \\ &= ((g_1g'_1, g_2g'_2), g_3g'_3). \end{aligned}$$

Hence, this is an isomorphism, as required.

4. Which of the following direct product groups are isomorphic? Justify your answer.

$$C_{12}, \quad C_6 \times C_2, \quad C_3 \times C_4, \quad C_3 \times C_2 \times C_2.$$

Solution: By the result in the lecture, we know that $C_n \times C_m \cong C_{nm}$ if and only if n, m are coprime. Therefore, C_{12} is not isomorphic to $C_6 \times C_2$, but $C_{12} \cong C_3 \times C_4$. We also note that $C_6 \times C_2 \cong (C_3 \times C_2) \times C_2$, since 2, 3 are coprime. Hence partitioning the groups into sets which are mutually isomorphic, we have:

$$\{C_{12}, C_3 \times C_4\}, \quad \{C_6 \times C_2, C_3 \times C_2 \times C_2\}.$$

5. Suppose that G_1, G_2 are groups.

- (a) Let $H_1 \leq G_1, H_2 \leq G_2$ be subgroups of G_1, G_2 respectively. Show that $H_1 \times H_2$ is a subgroup of $G_1 \times G_2$.
- (b) Is every subgroup of $G_1 \times G_2$ of the form $H_1 \times H_2$, where $H_1 \leq G_1, H_2 \leq G_2$? Give a proof or counterexample as appropriate.

Solution:

(a) We verify each of the group axioms:

(G1) *Closure.* Let $(h_1, h_2), (h'_1, h'_2) \in H_1 \times H_2$. Then:

$$(h_1, h_2)(h'_1, h'_2) = (h_1 h'_1, h_2 h'_2) \in H_1 \times H_2,$$

since $h_1 h'_1 \in H_1$ and $h_2 h'_2 \in H_2$ by closure of the subgroups H_1, H_2 respectively.

(G2) *Associativity.* $H_1 \times H_2$ inherits associativity from the larger group $G_1 \times G_2$.

(G3) *Identity.* The identity is contained in $H_1 \times H_2$, since the identities $e_1 \in H_1$ and $e_2 \in H_2$ must be in their respective subgroups, implying $(e_1, e_2) \in H_1 \times H_2$.

(G4) *Inverses.* If $h_1 \in H_1$ and $h_2 \in H_2$, then we have $h_1^{-1} \in H_1$ and $h_2^{-1} \in H_2$ by the axiom of inverses for these subgroups. But this implies that $(h_1^{-1}, h_2^{-1}) \in H_1 \times H_2$, which is the inverse of the element (h_1, h_2) .

(b) No, not every subgroup is of this form. Consider $C_2 \times C_2 = \{(e, e), (e, r), (r, e), (r, r)\}$. The only subgroups of C_2 are $\{e\}$ and C_2 , hence the subgroups suggested by part (a) are:

$$\{e\} \times \{e\} = \{(e, e)\}, \quad \{e\} \times C_2 = \{(e, e), (e, r)\}, \quad C_2 \times \{e\} = \{(e, e), (r, e)\}, \quad C_2 \times C_2.$$

But there is also $\{(e, e), (r, r)\}$ as a subgroup (since (r, r) has order two), which is not equal to any of these.

The direct product theorem

6. Let H, K be the groups of matrices, under matrix multiplication, generated by the sets:

$$\left\{ \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} \right\}, \quad \left\{ \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \right\}$$

respectively. Let G be the smallest group of matrices containing both H, K as subgroups. Show that $G \cong H \times K$.

Solution: Define:

$$A = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Then $A^2 = I$ and $B^2 = I$, so that $H = \{I, A\}$ and $K = \{I, B\}$. Further, we note that $AB = BA$, so all elements of a group generated by the elements A, B can be written in the form:

$$A^\alpha B^\beta,$$

using the fact that the matrices A, B commute. If $A^\alpha B^\beta = A^{\alpha'} B^{\beta'}$, then we have $A^{\alpha-\alpha'} = B^{\beta'-\beta}$, which can only occur if $\alpha = \alpha'$ and $\beta = \beta'$ (modulo two). Thus every element of the group can be uniquely expressed in this form. Further, $A^\alpha B^\beta = B^\beta A^\alpha$ for each $\alpha = 0, 1, \beta = 0, 1$, so the elements of the subgroups H, K are commuting. Thus we have:

$$G \cong H \times K,$$

by the hypotheses of the direct product theorem.

7. Using the direct product theorem, show that $\mathbb{Z}_{35}^* \cong C_4 \times C_6$. [Hint: consider the subgroups generated by 8 and 31.]

Solution: This is very similar to the lecture. We note that:

$$\mathbb{Z}_{35}^* = \{1, 2, 3, 4, 6, 8, 9, 11, 12, 13, 16, 17, 18, 19, 22, 23, 24, 26, 27, 29, 31, 32, 33, 34\}.$$

Using the hint, we consider the subgroup H generated by 22:

$$8^2 = 29, \quad 8^3 = (-6) \cdot 8 = 22, \quad 8^4 = 8 \cdot 22 = 1.$$

We also consider the subgroup K generated by 31:

$$31^2 = (-4)^2 = 16, \quad 31^3 = 16 \cdot (-4) = 6, \quad 31^4 = 6 \cdot (-4) = 11,$$

$$31^5 = 11 \cdot (-4) = 26, \quad 31^6 = (-9) \cdot (-4) = 1.$$

Therefore $|H| = 4$ and $|K| = 6$, so $|H||K| = 24$, $H \cap K = \{1\}$, and all elements of H commute with all elements of K . It follows that $\mathbb{Z}_{35}^* \cong H \times K \cong C_4 \times C_6$, as required.

8. Using the direct product theorem, provide an alternative proof that $C_n \times C_m \cong C_{nm}$ if and only if n, m share no common factors except one.

Solution: First assume that n, m are coprime. Let $C_{nm} = \{e, r, r^2, \dots, r^{nm-1}\}$. Recall that C_{nm} has unique subgroups H, K isomorphic to C_n, C_m respectively, given by:

$$C_n \cong \{e, r^m, r^{2m}, \dots, r^{m(n-1)}\}, \quad C_m \cong \{e, r^n, r^{2n}, \dots, r^{n(m-1)}\}.$$

Certainly every element in the first subgroup commutes with every element in the second subgroup, because the group is Abelian. Note also that $|C_n| = n$ and $|C_m| = m$. Finally, observe that if $pn = qm$ for some p, q , we would have m dividing p and n dividing q , since m, n share no common factors except one. Thus $C_n \cap C_m = \{e\}$. By the direct product theorem then, we have $C_{nm} \cong C_n \times C_m$.

Now suppose that n, m are not coprime; then the lowest common multiple of n, m is some $k < nm$. Write $k = n\alpha, k = m\beta$ for some $\alpha < m, \beta < n$. The converse of the direct product theorem says that if C_{nm} is expressible as a direct product $C_n \times C_m$, then the subgroups *must* be those given above (since the cyclic group C_{nm} has unique subgroups of each size). But then r^k is in both subgroups, which contradicts the requirement that the subgroups intersect only in the identity. Thus there is no decomposition of this form.

9. Show that the full cubic group is isomorphic to $C_2 \times O$, where O is the chiral cubic group. [Hint: $-I$, where I is the identity, is a symmetry of the cube.]

Solution: Using the hint, we note that $-I$ is a symmetry of the cube (corresponding to ‘inverting’ the cube, swapping opposite vertices).

We have already seen that $|\{I, -I\}| = 2, |O| = 24$, so $|C_2||O| = 48$. Further $\{I, -I\} \cap O = \{I\}$, since the inversion $-I$ is not a pure rotation, as it changes the handedness of the cube. Finally, inversion can be represented as the matrix:

$$\begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix},$$

if the cube is centred on the origin and is axes-aligned. This clearly commutes with all other transformations (since it is a scalar multiple of the identity matrix). Thus $\{I, -I\}$ is completely commuting with O . By the direct product theorem, the full cubic group is therefore isomorphic to $C_2 \times O$, as required.

Classification of finite Abelian groups

10. List all Abelian groups of order 108.

Solution: Since $108 = 2^2 \cdot 3^3$, the possible non-isomorphic groups are:

$$\begin{array}{lll} C_{108} \cong C_4 \times C_{27}, & C_2 \times C_2 \times C_{27}, & C_4 \times C_3 \times C_9, \\ C_2 \times C_2 \times C_3 \times C_9, & C_4 \times C_3 \times C_3 \times C_3, & C_2 \times C_2 \times C_3 \times C_3 \times C_3. \end{array}$$

11. A finite Abelian group has order 16, and contains an element of order 8, but no elements of order 16. What is the group?

Solution: The group must be one of the C_{16} , $C_8 \times C_2$, $C_4 \times C_2 \times C_2$ and $C_2 \times C_2 \times C_2 \times C_2$. It has no element of order 16, so cannot be C_{16} . It contains an element of order eight, but all elements in $C_4 \times C_2 \times C_2$ and $C_2 \times C_2 \times C_2 \times C_2$ have order at most four. Thus the group is $C_8 \times C_2$.

Classification of groups

12. (a) A group G is called *decomposable* if it is isomorphic to an external direct product:

$$G \cong H \times K,$$

where H, K are non-trivial groups; a group is called *indecomposable* otherwise. Prove that every finite group can be written in the form:

$$G \cong G_1 \times G_2 \times \dots \times G_n,$$

where G_i is an indecomposable group. [Well beyond the scope of this course, the Krull-Schmidt theorem states that these indecomposable factors G_i are unique up to isomorphism and reordering.]

Solution: We proceed by induction. First, observe that every group of order 1 is isomorphic to C_1 , which is trivially indecomposable. Thus all groups of order one can be written in this form.

Suppose that all groups of order less than or equal to k can be written in the give form. Consider a group G of order $k + 1$. If G is indecomposable, we are done. Else, there exist non-trivial groups H, K such that $G \cong H \times K$. But then H, K have order less than or equal to k , since they are non-trivial. It follows that H, K are expressible in the forms $H \cong H_1 \times \dots \times H_\alpha$, $K \cong K_1 \times \dots \times K_\beta$, for some indecomposable factors. Thus G is also expressible as the direct product of indecomposable factors. Hence we're done by induction.

- (b) Show that that Q_8 is indecomposable.

Solution: If Q_8 were decomposable, we would be able to write $Q_8 \cong H \times K$ where H, K are non-trivial subgroups of Q_8 . They would necessarily have sizes 2 and 4 respectively. But the subgroups of Q_8 of order four are all isomorphic to C_4 ; hence the direct product would have to be isomorphic to $C_2 \times C_4$ which is Abelian. This is a contradiction, so Q_8 is indecomposable.

(c) Let $r^\alpha m^\beta, r^{\alpha'} m^{\beta'} \in D_n$. Show that:

$$(r^\alpha m^\beta)(r^{\alpha'} m^{\beta'}) = (r^{\alpha'} m^{\beta'})(r^\alpha m^\beta)$$

if and only if $\alpha + (-1)^\beta \alpha' = \alpha' + (-1)^{\beta'} \alpha$ modulo n . Deduce that:

- rotations always commute;
- a rotation r^α and a reflection $r^{\alpha'} m$ commute if and only if $2\alpha = 0$ modulo n ;
- two reflections $r^\alpha m$ and $r^{\alpha'} m$ commute if and only if $2(\alpha - \alpha') = 0$ modulo n .

Hence, argue that D_n is decomposable if n is twice an odd number, else it is indecomposable.

Solution: Using the multiplication rule in D_n , the given equation holds if and only if:

$$r^{\alpha+\alpha'(-1)^\beta} m^{\beta+\beta'} = r^{\alpha'+\alpha(-1)^{\beta'}} m^{\beta'+\beta},$$

which holds if and only if $\alpha + \alpha'(-1)^\beta = \alpha' + \alpha(-1)^{\beta'}$ modulo n , as required. Examining the cases in the question then, we note:

- Rotations have $\beta = \beta' = 0$. Hence the elements $r^\alpha, r^{\alpha'}$ commute, corresponding to the fact that $\alpha + \alpha' = \alpha' + \alpha$ modulo n .
- If we have one reflection and one rotation, we take $\beta = 0, \beta' = 1$. Then the elements $r^\alpha, r^{\alpha'} m$ commute if and only if $\alpha + \alpha' = \alpha' - \alpha$ modulo n , corresponding to requiring $2\alpha = 0$ modulo n . Thus the only non-identity rotation that commutes with a reflection is $r^{n/2}$, if n is even, else there is no such rotation (indeed $r^{n/2}$ commutes with everything).
- If we have two reflections, we take $\beta = \beta' = 1$. Then the elements $r^\alpha m, r^{\alpha'} m$ commute if and only if $\alpha - \alpha' = \alpha' - \alpha$ modulo n , corresponding to requiring $2(\alpha - \alpha') = 0$ modulo n . If n is odd, then 2 is invertible modulo n , implying that $\alpha = \alpha'$. Thus no reflections commute when n is odd. If n is even, then $\alpha = \alpha' + n/2$, so that reflections that are at an angle $\pi/2$ commute.

Suppose now that $D_n \cong H \times K$, where H, K are subgroups of D_n . If neither H, K contain any reflections, then they cannot generate all of D_n , because the product of rotations is always a rotation. Thus we may assume without loss of generality that $r^k m \in H$ for some k . We now split into two cases:

- Suppose that K contains a reflection too, say $r^l m$ (which must be distinct, for $H \cap K = \{e\}$). We must have $r^l m$ and $r^k m$ commuting so we need $2(l - k) = 0$ modulo n . If n is odd, this cannot occur unless $l = k$ so K contains no reflections. On the other hand, if n is even, then $l = k + n/2$. This gives $\{e, r^k m\} \subseteq H$ and $\{e, r^{k+n/2} m\} \subseteq K$. Observe now that H cannot contain any other reflections, because they would not commute with $r^{k+n/2} m$. Further H cannot contain any rotations except $r^{n/2}$, else they would not commute with $r^{k+n/2} m$. If $r^{n/2} \in H$, then we would have $r^{n/2} r^k m = r^{k+n/2} m \in H$ by closure. This is a contradiction, so we require: $H = \{e, r^k m\}$ and $K = \{e, r^{k+n/2} m\}$. This implies the order of the group is 4 so that $n = 2$; we already know that $D_2 \cong C_2 \times C_2$, so we're done in this case.
- In the remaining case, K contains only the identity and rotations. If n is odd, there are no rotations commuting with the reflection $r^k m$, so K is trivial; contradiction. When n is even, the only rotation commuting with $r^k m$ is $r^{n/2}$. This implies that $K = \{e, r^{n/2}\}$. Thus, using our classification of the subgroups of D_n , the other factor must be one of the subgroups isomorphic to $D_{n/2}$, of order n . The only possibilities are:

$$H = \{e, r^2, r^4, \dots, r^{n-2}, m, r^2 m, r^4 m, \dots, r^{n-2} m\},$$

or:

$$H' = \{e, r^2, r^4, \dots, r^{n-2}, rm, r^3m, \dots, r^{n-1}m\}.$$

If $n/2$ is even, then $H \cap K$ and $H' \cap K$ are both not the identity, so we cannot have a direct product decomposition. Otherwise, if $n/2$ is odd, we have $|H||K| = 2n$, $H \cap K = \{e\}$ and all elements of H commuting with all elements of K . Thus $D_n \cong H \times K$, as required.

- (d) Using parts (b) and (c), make a complete list of indecomposable groups up to and including order 11.

Solution: The indecomposable groups of each order are:

- C_1 ;
- C_2 ;
- C_3 ;
- C_4 (since $C_2 \times C_2$ is obviously decomposable);
- C_5 ;
- D_3 (since $C_6 \cong C_2 \times C_3$ is obviously decomposable);
- C_7 ;
- D_4 , Q_8 and C_8 (since $C_4 \times C_2$ and $C_2 \times C_2 \times C_2$ are obviously decomposable);
- C_9 (since $C_3 \times C_3$ is obviously decomposable);
- D_5 (since $C_{10} \cong C_2 \times C_5$ is obviously decomposable);
- C_{11} .

13. Is it true that if a group's non-identity elements all have order three, then the group must be Abelian? Provide a proof or counterexample as appropriate.

Solution: No, this is false. Consider the group consisting of matrices of the form:

$$\begin{pmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{pmatrix},$$

where a, b, c are taken modulo 3. This is a group, following the same argument as Examples Sheet 3, Question 6. It is non-Abelian since for example:

$$\begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 1 & 2 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{pmatrix},$$

but:

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{pmatrix}.$$

Finally, observe that every non-identity element has order three. The group has order 27 (three choices for a, b, c each), so all elements have order one, three, nine or 27. We note that:

$$\begin{pmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{pmatrix}^2 = \begin{pmatrix} 1 & 2a & 2b + ac \\ 0 & 1 & 2c \\ 0 & 0 & 1 \end{pmatrix},$$

so that:

$$\begin{pmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{pmatrix}^3 = \begin{pmatrix} 1 & 3a & 3b + 3ac \\ 0 & 1 & 3c \\ 0 & 0 & 1 \end{pmatrix} = I,$$

since all elements are taken modulo 3.